BYTE.ME: FOOD DELIVERY APP

Software project documentation · AFFIAT · Afro-Asian Forum for Innovation and Technology · Aswan Finals 2026



1. Project Overview | نظرة عامة على المشروع

Field	Value
Project Title	BYTE.ME: Food Delivery App
Project Type	Web application
Description	BYTE.ME is a food delivery website, like Talabat. You pick a restaurant, look at its dishes with photos and prices, put them in your cart and place an order. Restaurant owners and admins add restaurants and dishes and move each order from pending to completed.
Target Users	Hungry customers who want to order food, restaurant owners who want to sell online, and admins who look after the menus and orders. Guests can look around before they sign up.
Live app	https://byte-me.streamlit.app/register

Tech Stack | التقنيات المستخدمة

Category	Technology	Purpose
Language	Python	All the app's logic
Framework	Streamlit 1.64	Web pages, forms, the role-based menu and session state
Database	SQLite (built into Python)	Users, restaurants, dishes, orders and payments in `database/database.db`

Category	Technology	Purpose
Styling	HTML + CSS inside `st.markdown`	The orange banner, cards and dish photos, defined once in `app.py`
Images	Web links (Wikimedia Commons, TheMealDB)	Dish photos saved in `products.image_url`

Dependencies | المكتبات

Package	Version	Purpose
streamlit	1.64.0	The whole web interface. It is the only package in `requirements.txt`.
sqlite3	Standard library	Talking to the database. Nothing to install.
base64	Standard library	Turning `images/background.png` into text for the CSS.

2. Problem Statement | المشكلة

Ordering food by phone is slow. The line is often busy, you can't see the menu or the prices, and after you hang up you don't know what is happening with your order. Hungry people in Egyptian cities face this every day, and so do small restaurants that can't build an app of their own.

Why does it matter? | لماذا هذا مهم؟

When ordering is hard, people wait longer, get the wrong dish or give up and eat something else. Small restaurants lose customers to big chains just because they are harder to order from.

How is it currently solved? | كيف تُحل المشكلة حاليًا؟

Most people call the restaurant or use a big delivery app. Calling means busy lines and no menu in front of you. Big apps work well, but a small local restaurant has to pay and go through a long sign-up to join them.

3. Technical Architecture | البنية التقنية

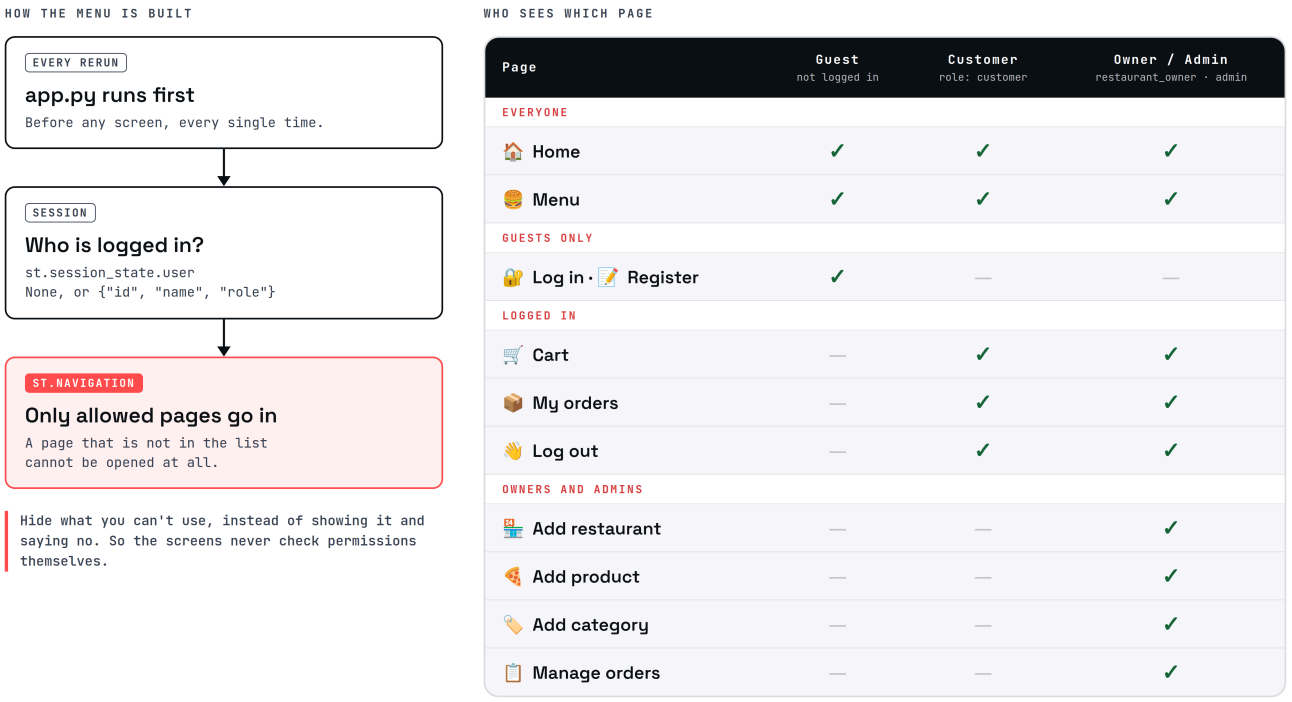
Project Structure | هيكل المشروع

```
adel-byte-me/
├── app.py                      # Runs first every time: look of the app, cart/user
defaults, role-based menu
├── app_pages/
│   ├── home.py                 # Restaurant list with search and food-type filter
│   ├── menu.py                 # One restaurant's dishes, add to cart
│   ├── login.py                # Log in with email and password
│   ├── register.py             # Create a customer account
│   ├── cart.py                 # Cart, delivery details, payment, place order
│   ├── orders.py               # The logged-in user's orders
│   ├── add_restaurant.py       # Owner/admin: add a restaurant
│   ├── add_product.py          # Owner/admin: add a dish
│   ├── add_category.py         # Owner/admin: add a category
│   ├── manage_orders.py        # Owner/admin: change each order's status
│   └── logout.py               # Clear the session and go back to Home
├── database/
│   ├── create_tables.sql       # The schema (8 tables)
│   ├── insert_*.SQL            # Demo data, loaded in foreign-key order
│   ├── add_product_photos.SQL # Photo links for the 16 demo dishes
│   └── database.db             # The SQLite database the app uses
├── images/                     # logo.png, favicon.png, background.png
├── docs/                       # This documentation, the poster, the slides and the
diagrams
└── requirements.txt            # streamlit==1.64.0
```

Architecture and Data Flow | تدفق البيانات

```
User's browser
    → app.py (runs before every page)
        → reads st.session_state.user and builds the menu with st.navigation
        → runs the chosen screen in app_pages/
            → the screen opens its own sqlite3 connection
            → plain SQL: SELECT / INSERT / UPDATE
            → commit and close
```

There is no services layer. Each screen holds its own SQL so it can be read on its own, top to bottom.



Key Files | الملفات الأساسية

File	Purpose
`app.py`	Sets the page title and icon, the logo, the shared CSS and the marble background. Decides which pages go in the menu for guests, customers, and owners/admins.
`app_pages/cart.py`	The biggest screen. Shows the cart with ➖ / ➕, takes the delivery details and payment method, checks stock again, and saves the order in five steps.
`app_pages/menu.py`	Shows dishes that are available and in stock. Stops the cart from mixing restaurants or going over the stock.
`app_pages/manage_orders.py`	Lists every order with its customer, address and payment, and lets an owner or admin change the status.
`database/create_tables.sql`	The 8 tables, with `CHECK` rules for statuses, roles and payment methods.

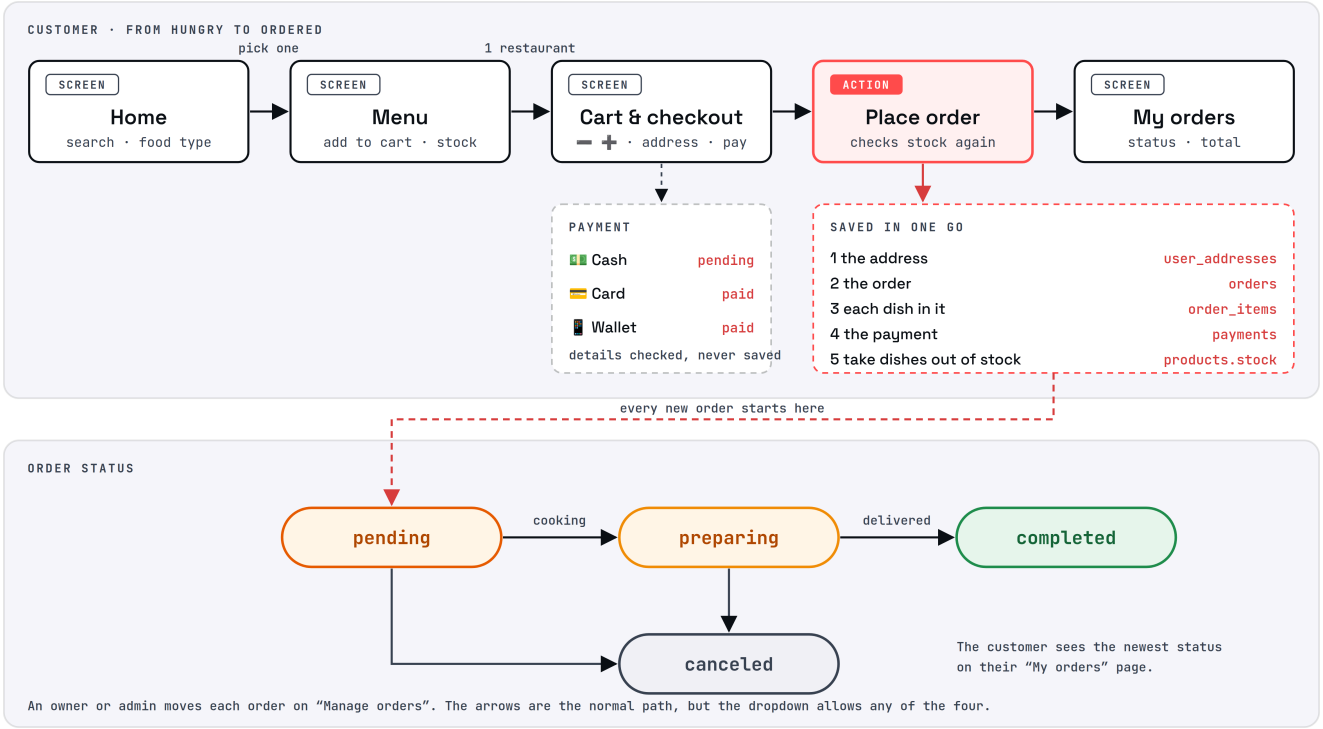
4. Features & User Flow | المميزات ورحلة المستخدم

Feature List | قائمة المميزات

#	Feature	Description	Status
1	Restaurant list	Search by name and filter by food type, three cards per row.	Complete

#	Feature	Description	Status
2	Menus with photos	Each dish shows a photo, description, rating, calories and price.	Complete
3	Accounts	Register as a customer and log in with email and password.	Complete. Passwords are plain text (see Limits).
4	Cart	➖ / ➕ / 🗑️ for each dish, up to the stock, from one restaurant only.	Complete
5	Checkout	City, address, phone, delivery notes, and cash, card or mobile wallet.	Complete. No real payments.
6	My orders	Each order's number, status, total, payment and notes.	Complete
7	Owner/admin tools	Add restaurants, dishes and categories.	Complete. Owners can edit any restaurant.
8	Manage orders	Move an order between pending, preparing, completed and canceled.	Complete

User Flow | رحلة المستخدم



```
Home (app_pages/home.py)
    → See menu → Menu (app_pages/menu.py)
        → guest: "Log in to order" → Log in / Register
        → logged in: Add to cart
            → Cart (app_pages/cart.py)
                → delivery details + payment → Place order
                    → saves address, order, items, payment; takes dishes out of stock
                    → My orders (app_pages/orders.py)

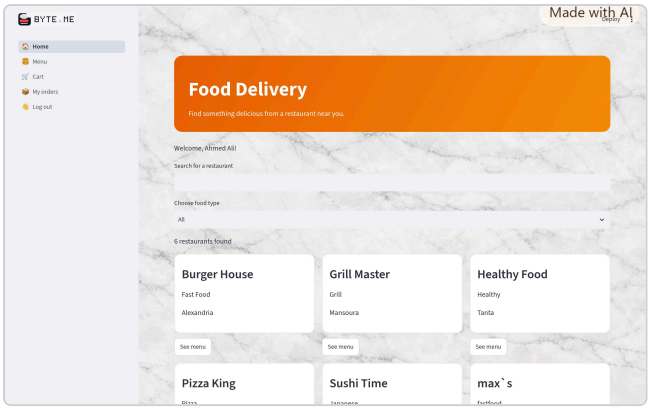
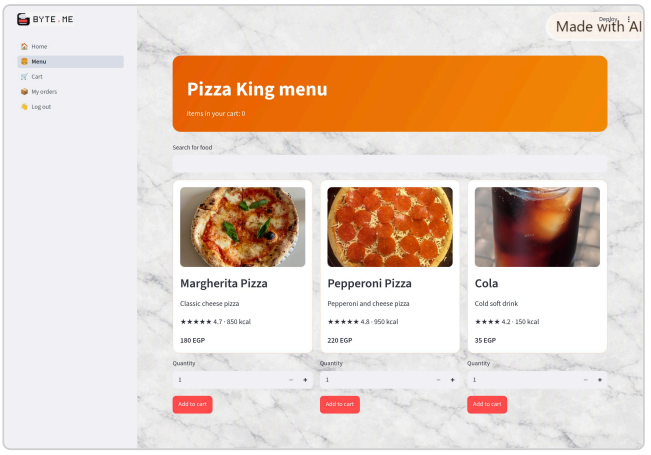
Owner / admin:
    Manage orders (app_pages/manage_orders.py) → pending → preparing → completed (or
canceled)
    Add restaurant / Add product / Add category
```

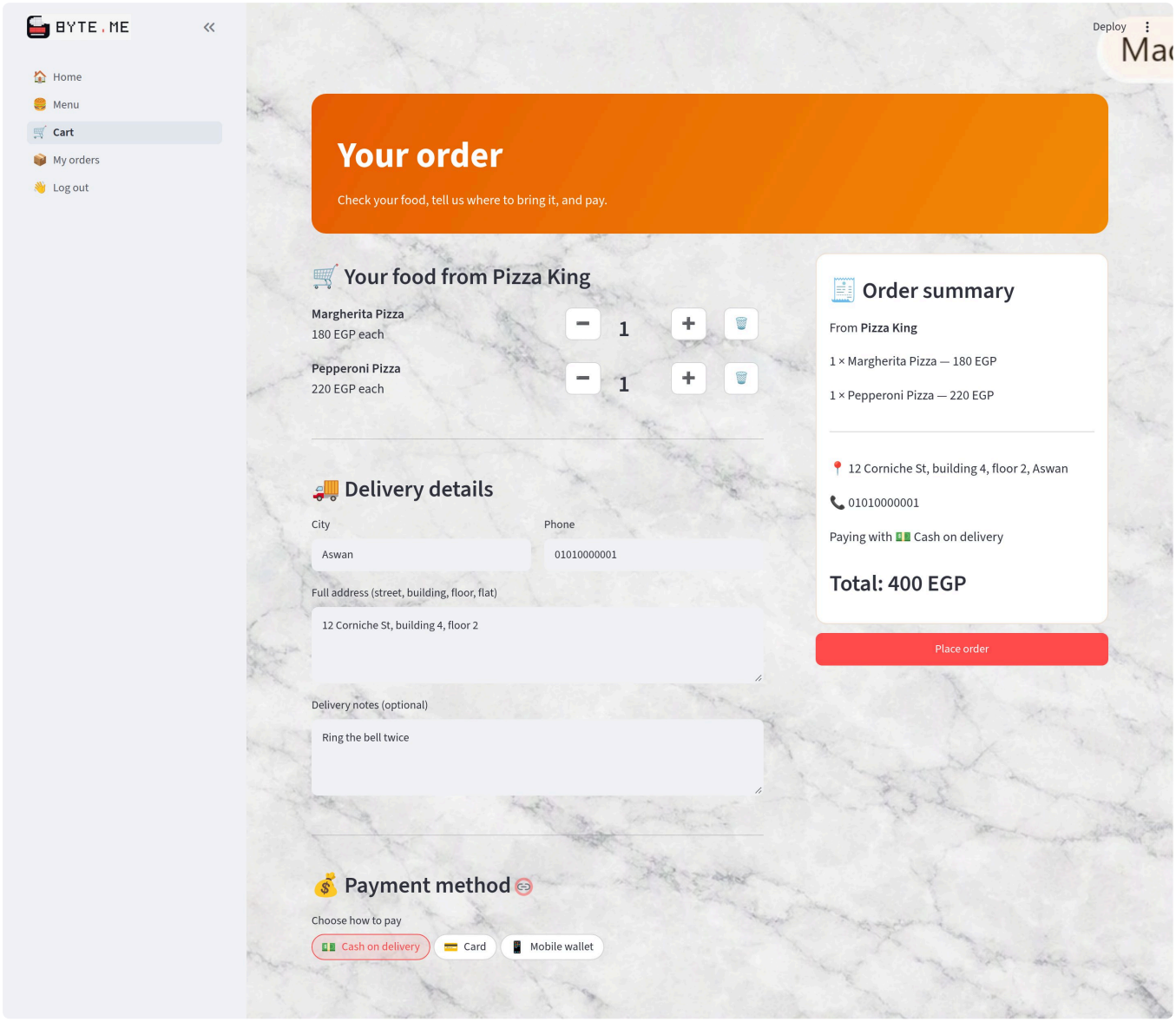
Screen Reference | الشاشات

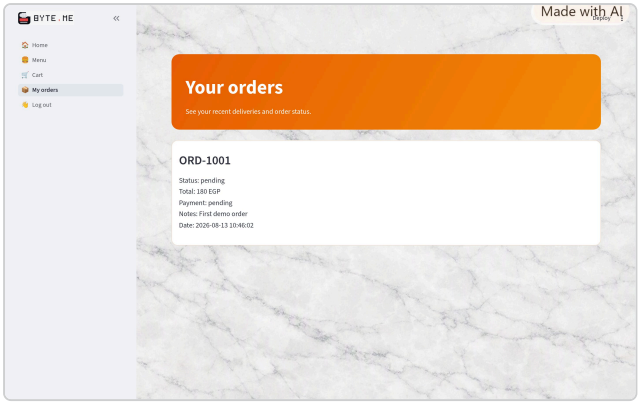
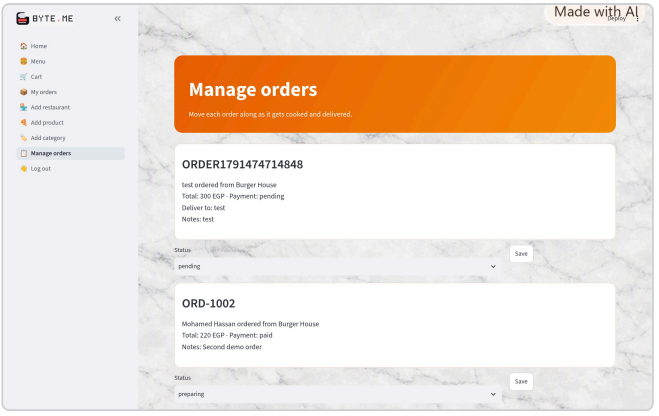
Screen	File	Who sees it	Description
Home	`app_pages/home.py`	Everyone	Welcome line, search box, food-type dropdown, restaurant cards.
Menu	`app_pages/menu.py`	Everyone	The chosen restaurant's dishes. Logged-in users get a quantity box and "Add to cart".
Log in	`app_pages/login.py`	Guests	Email and password.
Register	`app_pages/register.py`	Guests	Name, phone, age, password, address, date of birth, email.
Cart	`app_pages/cart.py`	Logged in	Cart, delivery details, payment method, order summary, "Place order".
My orders	`app_pages/orders.py`	Logged in	The user's orders, newest first.
Add restaurant / product / category	`app_pages/add_*.py`	Owners, admins	Forms that add a row to the database.
Manage orders	`app_pages/manage_orders.py`	Owners, admins	Every order with a status dropdown and "Save".
Log out	`app_pages/logout.py`	Logged in	Clears the session and reloads.

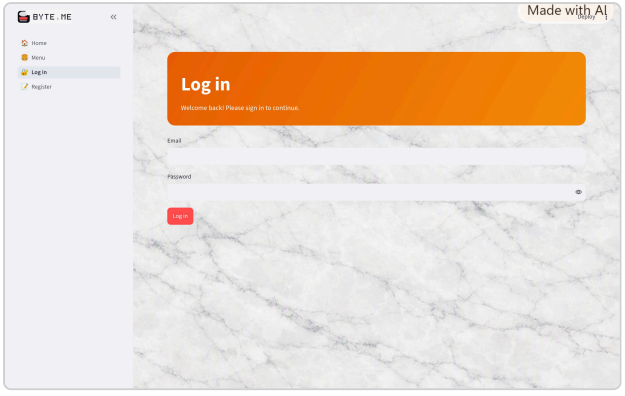
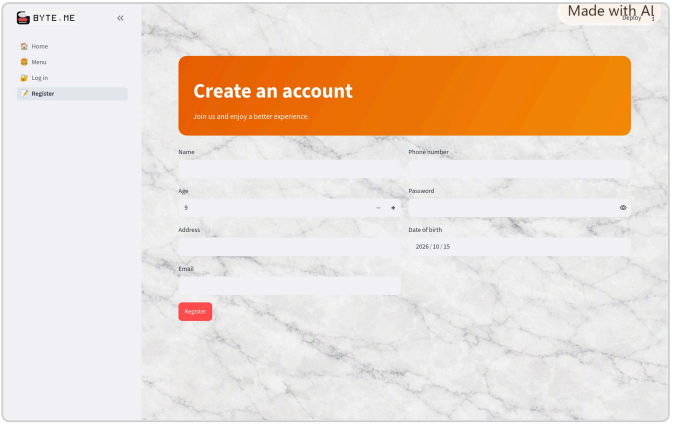
5. UI/UX Design | تصميم واجهة المستخدم

Screenshots | لقطات الشاشة

Home	Menu
	



My orders	Manage orders
	

Log in	Register
	

Try the app: https://byte-me.streamlit.app/register

Design System | نظام التصميم

Element	Current implementation
Logo	A pixel-art lunch box with a bite taken out of the corner, next to "BYTE.ME" in pixel letters with a red dot (`images/logo.png`). The mark alone is the browser-tab icon (`images/favicon.png`).
Primary color	Orange banner gradient, `#E85D04` → `#F48C06`
Accent color	Logo red `#FF4B4C`, also used by Streamlit for primary buttons
Ink / background	Logo ink `#0E1118` on a white marble background (`images/background.png`)

Element	Current implementation
Cards	White, `1px` border `#F0DFCC`, `14px` rounded corners
Dish photos	A `div` with the photo as its background, all `180px` tall so cards line up
Font	Streamlit's default font in the app. The poster and slides use Silkscreen (pixel), Space Grotesk and JetBrains Mono.
Buttons	One red `type="primary"` button per screen for the main action
Layout	Wide pages, three cards per row, a left menu that changes with the user's role

6. Data Sources | مصادر البيانات

Data Sources | مصادر البيانات

Source	Type	Description
Sign-up form	User input	Name, age, address, email, phone, password and date of birth go into `users`.
Checkout form	User input	City, address and phone go into `user_addresses`. Delivery notes go into `orders.notes`. Card and wallet details are checked but never saved.
Owner/admin forms	User input	New restaurants, dishes and categories.
`database/insert_*.SQL`	Demo data	5 restaurants, 8 categories, 6 dishes, 3 users and 2 orders. The `database.db` in the repo has more, including the 16 dishes that `add_product_photos.SQL` gives photos to.
Wikimedia Commons, TheMealDB	Web images	Dish photos, linked by URL in `products.image_url`.

Database Schema | قاعدة البيانات

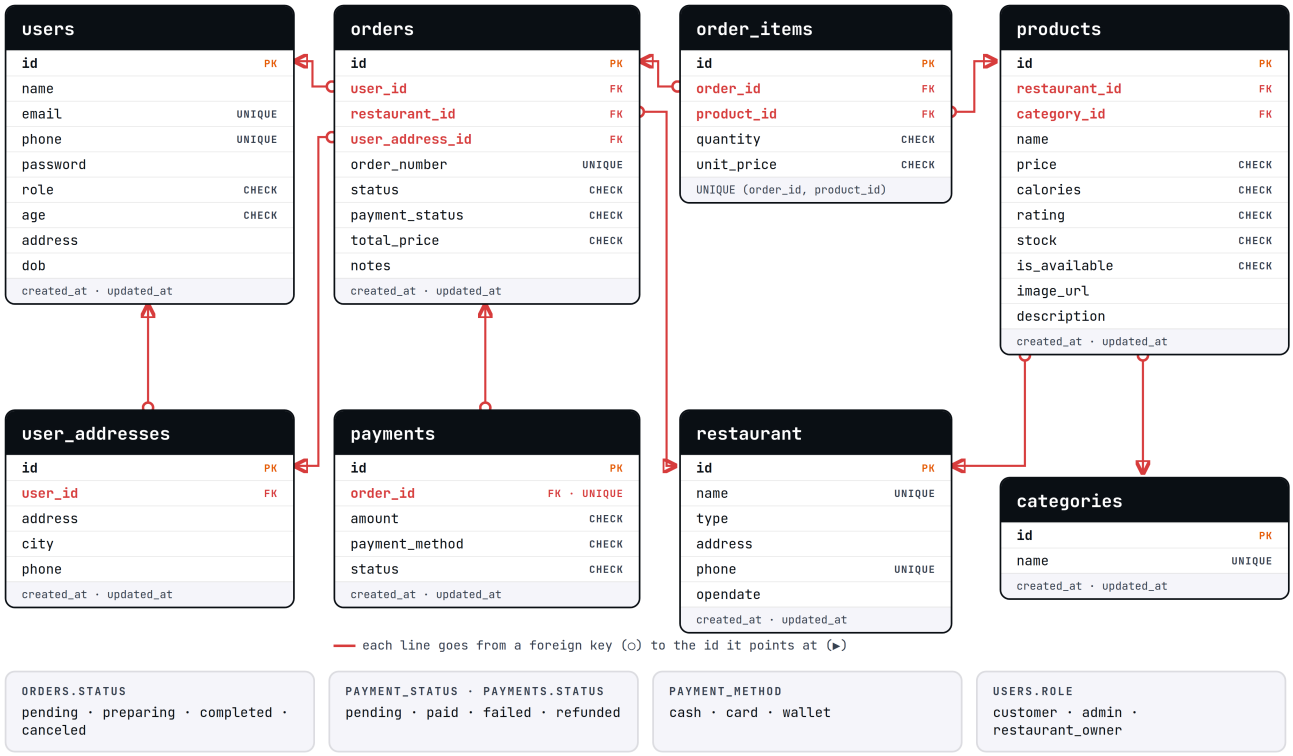


Table	Columns	Description
`users`	`id`, `name`, `age`, `address`, `email`, `phone`, `password`, `role`, `dob`	Accounts. `role` is customer, admin or restaurant_owner.
`restaurant`	`id`, `name`, `opendate`, `address`, `type`, `phone`	Restaurants. `type` is the food type used by the Home filter.
`categories`	`id`, `name`	Dish categories.
`products`	`id`, `restaurant_id`, `category_id`, `name`, `calories`, `price`, `rating`, `description`, `image_url`, `stock`, `is_available`	Dishes.
`user_addresses`	`id`, `user_id`, `address`, `city`, `phone`	One new row for each order's delivery address.
`orders`	`id`, `user_id`, `restaurant_id`, `user_address_id`, `order_number`, `status`, `notes`, `total_price`, `payment_status`	Orders. `order_number` comes from the time in milliseconds.
`order_items`	`id`, `order_id`, `product_id`, `quantity`, `unit_price`	The dishes in each order.
`payments`	`id`, `order_id`, `amount`, `payment_method`, `status`	One payment per order: cash, card or wallet.

Every table except `order_items` also has `created_at` and `updated_at`. `order_items` only has `created_at`.

External APIs | الخدمات الخارجية

API	Purpose	Rate Limit
None	The app calls no outside API. Dish photos are plain image links that the browser loads.	Not applicable

Limits | حدود المشروع

These shortcuts were taken on purpose to keep the code simple. The full list is in the README.

- Passwords are saved as plain text.
- Roles are set by hand in the database. Everyone who signs up is a customer.
- Restaurant owners can do everything admins can, for every restaurant.
- There are no real payments. Card and wallet orders are marked "paid" at once, and cash stays "pending".
- Reloading the page logs you out and empties the cart.
- Orders and sign-ups on the live app are lost when it restarts or is redeployed.

7. Educational Content | المحتوى التعليمي

Learning Objectives | أهداف التعلم

Through building this project, a student can practice:

1. Building a web app in pure Python with Streamlit: forms, buttons, columns, and `st.session_state` to remember who is logged in and what is in the cart.
2. Designing a database with 8 tables linked by ids, and reading and writing it with plain SQL (`SELECT`, `INSERT`, `UPDATE`).
3. Logging people in and showing each role (guest, customer, restaurant owner, admin) only the pages they are allowed to open, using `st.navigation`.
4. Checking what people type with plain `if`s: an 11-digit phone number, a 16-digit card number, an email that isn't already used, and enough stock for each dish.
5. Following one order through five database steps: save the address, the order, its dishes and the payment, then take the dishes out of the stock.

Cultural / Historical Context | السياق الثقافي

Topic	Description
Food delivery in Egypt	Apps like Talabat changed how people in Egyptian cities order food: many now order on their phone instead of calling. BYTE.ME is a small, simple version that shows how such an app works inside.

Topic	Description
Egyptian cities	The demo restaurants are in Cairo, Alexandria, Giza, Mansoura and Tanta, prices are in Egyptian pounds (EGP), and phone numbers must have 11 digits like Egyptian mobile numbers.
Paying safely	Many people in Egypt still pay cash on delivery, so BYTE.ME offers cash next to card and mobile wallet. It never saves card details, because keeping them safe needs real security that a school demo doesn't have.

Curriculum Alignment | الارتباط بالمنهج

Skill Area	What the Student Practices
Problem Solving	Splitting "order food online" into small screens: restaurant list, menu, cart, checkout and my orders.
Computational Thinking	Turning an order into data: rows in tables linked by ids, with a status that moves step by step from pending to completed.
Creativity	Designing the BYTE.ME name, the pixel-art logo with a bite taken out of it, and the banners and dish cards.
Collaboration	Built alone.
Technical Writing	Writing the README, this documentation and the Limits list so another student can set up and run the app.

8. Marketing Plan | خطة تسويقية

Target Audience | الجمهور المستهدف

- **Primary:** Hungry customers in Egyptian cities who order food on their phone or computer.
- **Secondary:** Small local restaurants that want to take orders online without building their own app.

Platforms | المنصات

- Phone and desktop web browsers, at https://byte-me.streamlit.app (nothing to install).
- The Afro-Asian Forum for Innovation and Technology (AFFIAT), Aswan Finals 2026, with a QR code on the poster and slides that opens the live app.

9. Student Worksheet | ورقة عمل الطالب

My Project Information | معلومات مشروعي

Student Name: Adel Ghonemy
Project Title: BYTE.ME: Food Delivery App

What is your project about? | عن ماذا يتحدث مشروعك؟

BYTE.ME is my food delivery website, like Talabat. You choose a restaurant, add dishes to your cart and place an order.

What problem does it solve? | ما المشكلة التي يحلها؟

Ordering food by phone is slow, and you can't see the menu or follow your order. BYTE.ME puts the menu, the cart and the order's status in one place.

My Data | بياناتي

Where does data come from?

Customers type in their own sign-up details, delivery address and orders. Restaurant owners and admins add restaurants, dishes and categories. The demo restaurants, dishes and photo links came from the SQL files in `database/`.

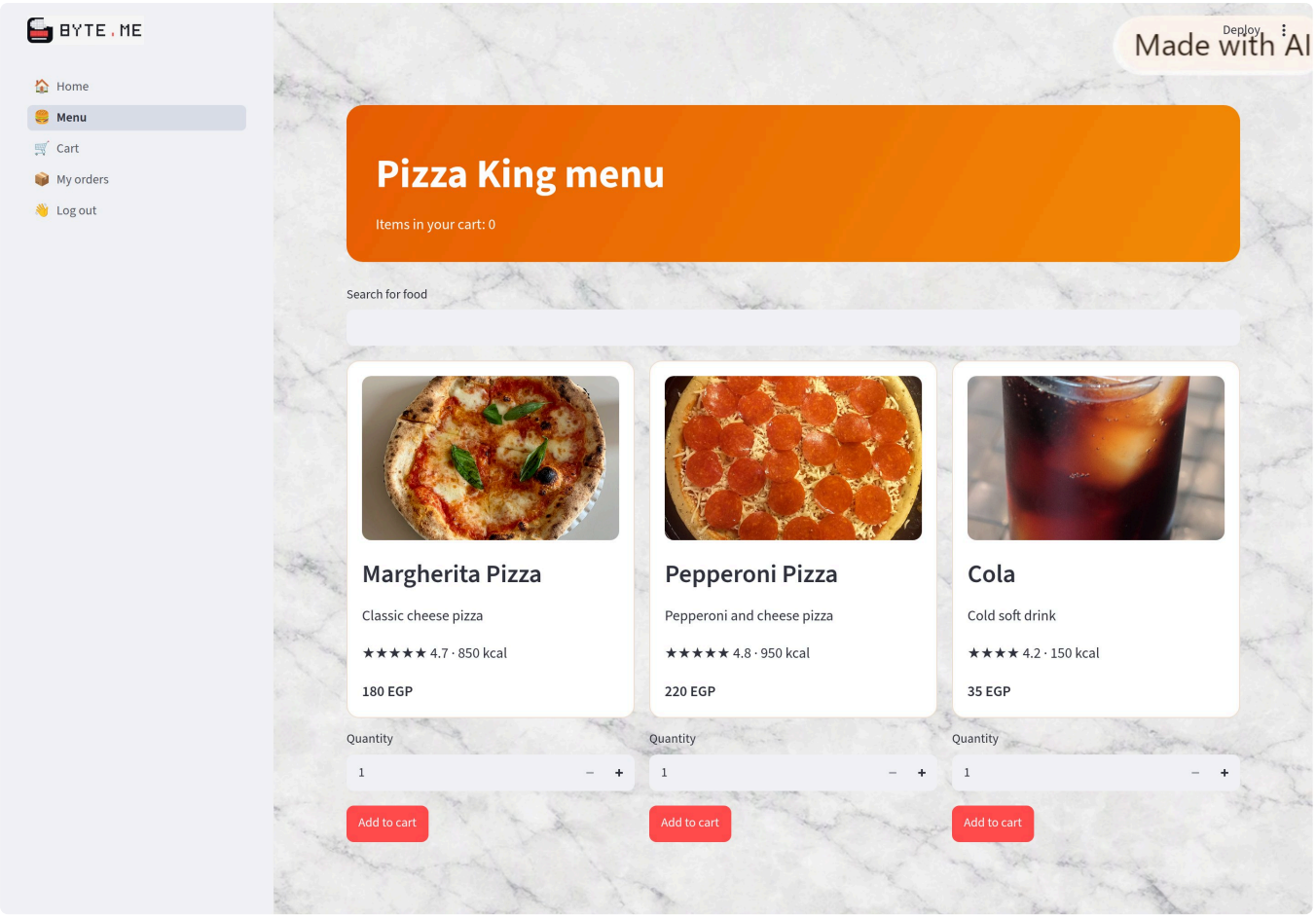
What data is stored?

Eight tables: `users`, `restaurant`, `categories`, `products`, `orders`, `order_items`, `payments` and `user_addresses`. They keep the accounts, the restaurants, the dishes with their price and stock, each order with its dishes and status, the payments and the delivery addresses.

What I Learned | ماذا تعلمت

1. **The hardest part:** The checkout. One button has to save the address, the order, its dishes and the payment, and take the dishes out of the stock, without mixing up anything.
2. **The most fun part:** Designing the BYTE.ME logo and seeing my dishes show up with real photos.
3. **If I had more time:** I would add real online payments, hide the passwords properly, and let each restaurant owner change only their own restaurant.
4. **One thing I would do differently:** I would plan all the database tables before writing the screens, so I wouldn't have to change them later.

Screenshot of My Project | لقطة من مشروعي



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